BCL1233

System Analysis and Design

Assignment 3
Logic and Data Modeling for a Hybrid Work Monitoring System

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Course: Bachelor of Computer Science (ODL)

Submission Date: July 2026

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Section 1: Logic Modeling (40 marks)

1.1 Analysis of Business Rules

The Hybrid Work Schedule Approval Process is governed by four distinct business rules. Each rule imposes a specific constraint that must be evaluated before a hybrid work request can be approved or rejected.

Rule	Description	Type
BR-01	Employees may work remotely for a maximum of three days per week.	Capacity constraint (individual)
BR-02	Requests must be submitted at least two days before the requested date.	Timing constraint
BR-03	Supervisor approval is required for all requests.	Authorization constraint
BR-04	Each department must maintain at least 50% on-site staff availability.	Capacity constraint (departmental)

A Decision Table was selected to model these rules because it exhaustively enumerates all condition combinations, eliminates ambiguity, and is particularly well-suited to scenarios where multiple binary conditions must be evaluated in parallel to determine an outcome.

1.2 Decision Table

Conditions

ID	Condition	Values
C1	Is the request submitted at least two days before the requested date?	Y / N

C2	Would approving this request keep the employee's remote work days at three or fewer per week?	Y / N
C3	Does the supervisor approve the request?	Y / N
C4	Would approving this request keep the department's on-site staffing at or above 50%?	Y / N

Actions

ID	Action
A1	Approve the request and update the employee's work schedule
A2	Reject the request and send a notification to the employee

Reduced Decision Table

	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5
Conditions					
C1: Submitted ≥ 2 days before?	N	Y	Y	Y	Y
C2: Remote days ≤ 3 per week?	—	N	Y	Y	Y
C3: Supervisor approves?	—	—	N	Y	Y
C4: Department ≥ 50% on-site?	—	—	—	N	Y
Actions					
A1: Approve and update schedule					X
A2: Reject and notify employee	X	X	X	X	

Note: "—" indicates that the condition is irrelevant (a "don't care" state) for that rule. **Rule Descriptions**

Rule	Scenario	Outcome
1	The employee failed to submit the request at least two days in advance. Regardless of all other conditions, the request cannot proceed.	Rejected
2	The request was submitted on time but would cause the employee to exceed three remote work days in the week.	Rejected
3	The request is valid under C1 and C2, but the supervisor does not approve it.	Rejected
4	The supervisor approved the request, but approving it would drop the department's on-site staffing below the 50% threshold.	Rejected

5	All four conditions are satisfied. The request is approved, the employee's schedule is updated, and a confirmation notification is sent.	Approved

1.3 Explanation of the Selected Logic Model

A decision table is a structured modeling technique that maps combinations of conditions to corresponding actions. Each column, or rule, represents a unique scenario defined by the truth values of the conditions. The technique is especially valuable in system analysis because it forces the analyst to consider every possible combination, revealing gaps or contradictions that might otherwise go unnoticed.

For the Hybrid Work Schedule Approval Process, the decision table captures the interplay between individual constraints (remote day limit), procedural constraints (advance notice), human judgment (supervisor approval), and organizational constraints (department staffing). The table clarifies that a request can be rejected at four distinct checkpoints, and only when all checks pass does approval occur. This explicit traceability makes the decision table a natural bridge between the business rules provided by stakeholders and the system logic that developers will implement.

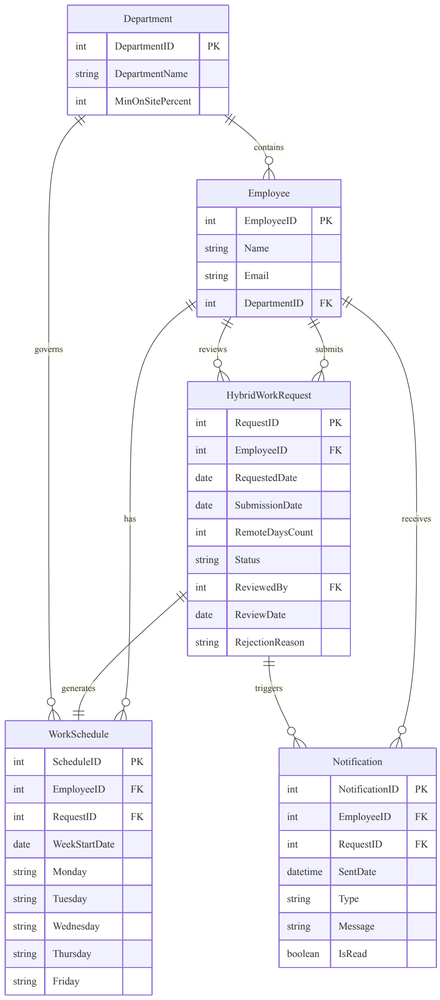
Section 2: Entity Relationship Diagram — ERD (35 marks)

2.1 Entity Identification

The following entities were identified by analysing the Hybrid Work Schedule Approval Process and its associated business rules:

Entity	Description
Employee	A staff member who may submit hybrid work requests and be assigned a work schedule
Department	An organisational unit with a mandated minimum on-site staffing percentage
HybridWorkRequest	A formal request submitted by an employee to work remotely on specific dates
WorkSchedule	The approved weekly schedule indicating the work location for each day
Notification	A record of a system-generated message sent to an employee regarding request status

2.2 Entity Relationship Diagram



2.3 Entity and Attribute Details

Department

Attribute	Type	PK/FK	Description
DepartmentID	INT	PK	Unique identifier for each department
DepartmentName	VARCHAR(100)	—	Name of the department
MinOnSitePercent	INT	—	Minimum percentage of staff that must be on-site (enforces BR-04)

Employee

Attribute	Type	PK/FK	Description
EmployeeID	INT	PK	Unique identifier for each employee
Name	VARCHAR(100)	—	Full name of the employee

Email	VARCHAR(150)	—	Corporate email address for notifications
DepartmentID	INT	FK	References the department the employee belongs to

HybridWorkRequest

Attribute	Type	PK/FK	Description
RequestID	INT	PK	Unique identifier for each request
EmployeeID	INT	FK	References the employee who submitted the request
RequestedDate	DATE	—	The specific date for which remote work is requested
SubmissionDate	DATE	—	The date the request was submitted (used to evaluate BR-02)
RemoteDaysCount	INT	—	Number of remote work days requested for that week (used to evaluate BR-01)
Status	VARCHAR(20)	—	Current status: Pending, Approved, or Rejected
ReviewedBy	INT	FK	References the EmployeeID of the supervisor who reviewed the request
ReviewDate	DATE	—	The date the supervisor made the approval decision
RejectionReason	VARCHAR(255)	—	Reason provided if the request is rejected

WorkSchedule

Attribute	Type	PK/FK	Description
ScheduleID	INT	PK	Unique identifier for each schedule record
EmployeeID	INT	FK	References the employee whose schedule this is
RequestID	INT	FK	References the approved request that generated this schedule
WeekStartDate	DATE	—	The Monday of the scheduled week
Monday	VARCHAR(10)	—	Work location for Monday (Office/Remote/Leave)
Tuesday	VARCHAR(10)	—	Work location for Tuesday
Wednesday	VARCHAR(10)	—	Work location for Wednesday
Thursday	VARCHAR(10)	—	Work location for Thursday
Friday	VARCHAR(10)	—	Work location for Friday

Notification

Attribute	Type	PK/FK	Description
NotificationID	INT	PK	Unique identifier for each notification
EmployeeID	INT	FK	References the employee who receives the notification
RequestID	INT	FK	References the request that triggered the notification
SentDate	DATETIME	—	Timestamp of when the notification was sent
Type	VARCHAR(20)	—	Notification type: Approved, Rejected, or Reminder
Message	VARCHAR(500)	—	Human-readable notification content
IsRead	BOOLEAN	—	Whether the employee has viewed the notification

2.4 Relationship Summary

Relationship	Parent	Child	Cardinality	Description
Contains	Department	Employee	1:M	One department contains many employees
Submits	Employee	HybridWorkRequest	1:M	One employee submits many requests
Reviews	Employee	HybridWorkRequest	1:M	One supervisor reviews many requests
Has	Employee	WorkSchedule	1:M	One employee has multiple weekly schedules
Receives	Employee	Notification	1:M	One employee receives many notifications
Generates	HybridWorkRequest	WorkSchedule	1:1	One approved request generates one schedule entry
Triggers	HybridWorkRequest	Notification	1:M	One request triggers one or more notifications
Governs	Department	WorkSchedule	1:M	One department's rules govern many schedule entries

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Section 3: Data Dictionary (15 marks)

3.1 Data Dictionary

Entity: Department

Attribute Name	Description	Data Type	Key Type	Validation Rules
DepartmentID	Unique identifier for each department	INT	PK	Auto-increment; not null
DepartmentName	Name of the department	VARCHAR(100)	—	Not null; must be unique; max 100 characters
MinOnSitePercent	Minimum percentage of staff that must work on-site	INT	—	Not null; value between 0 and 100; default 50

Entity: Employee

Attribute Name	Description	Data Type	Key Type	Validation Rules
EmployeeID	Unique identifier for each employee	INT	PK	Auto-increment; not null
Name	Full name of the employee	VARCHAR(100)	—	Not null; max 100 characters
Email	Corporate email address	VARCHAR(150)	—	Not null; must be unique; valid email format
DepartmentID	References the department the employee belongs to	INT	FK	Not null; must exist in Department table

Entity: HybridWorkRequest

Attribute Name	Description	Data Type	Key Type	Validation Rules
RequestID	Unique identifier for each request	INT	PK	Auto-increment; not null
EmployeeID	References the employee who submitted the request	INT	FK	Not null; must exist in Employee table
RequestedDate	The date for which remote work is requested	DATE	—	Not null; must be a future date

SubmissionDate	The date the request was submitted	DATE	—	Not null; must be on or before RequestedDate minus 2 days (BR-02)
RemoteDaysCount	Number of remote days requested in the same week	INT	—	Not null; must be between 1 and 3 (BR-01)
Status	Current status of the request	VARCHAR(20)	—	Not null; must be 'Pending', 'Approved', or 'Rejected'; default 'Pending'
ReviewedBy	References the supervisor who reviewed the request	INT	FK	Nullable until reviewed; must exist in Employee table
ReviewDate	Date the supervisor made the decision	DATE	—	Nullable until reviewed; must be on or after SubmissionDate
RejectionReason	Explanation if the request was rejected	VARCHAR(255)	—	Nullable; required only if Status = 'Rejected'

Entity: WorkSchedule

Attribute Name	Description	Data Type	Key Type	Validation Rules
ScheduleID	Unique identifier for each schedule record	INT	PK	Auto-increment; not null
EmployeeID	References the employee	INT	FK	Not null; must exist in Employee table
RequestID	References the approved request	INT	FK	Not null; must exist in HybridWorkRequest table
WeekStartDate	The Monday of the scheduled week	DATE	—	Not null; must be a Monday
Monday	Work location for Monday	VARCHAR(10)	—	Not null; must be 'Office', 'Remote', or 'Leave'
Tuesday	Work location for Tuesday	VARCHAR(10)	—	Not null; must be 'Office', 'Remote', or 'Leave'
Wednesday	Work location for Wednesday	VARCHAR(10)	—	Not null; must be 'Office', 'Remote', or 'Leave'
Thursday	Work location for Thursday	VARCHAR(10)	—	Not null; must be 'Office', 'Remote', or 'Leave'

Friday	Work location for Friday	VARCHAR(10)	—	Not null; must be 'Office', 'Remote', or 'Leave'

Entity: Notification

Attribute Name	Description	Data Type	Key Type	Validation Rules
NotificationID	Unique identifier for each notification	INT	PK	Auto-increment; not null
EmployeeID	References the recipient employee	INT	FK	Not null; must exist in Employee table
RequestID	References the triggering request	INT	FK	Not null; must exist in HybridWorkRequest table
SentDate	Timestamp of when the notification was sent	DATETIME	—	Not null; default current timestamp
Type	Category of the notification	VARCHAR(20)	—	Not null; must be 'Approved', 'Rejected', or 'Reminder'
Message	Content of the notification	VARCHAR(500)	—	Not null; max 500 characters
IsRead	Whether the notification has been viewed	BOOLEAN	—	Not null; default false

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Section 4: Model Validation and Integration (10 marks)

4.1 Role of the Logic Model in the Approval Process

The decision table defines the precise evaluation sequence governing every hybrid work request. Each of the four business rules maps directly to a condition in the table, and every possible combination is accounted for across the five rules. This eliminates ambiguity during implementation: a developer translating the table into code knows exactly which checks to perform, in what logical order, and what action to take for each outcome. Without the decision table, the interaction of timing constraints (BR-02), individual limits (BR-01), managerial judgement (BR-03), and departmental staffing rules (BR-04) would be open to subjective interpretation, increasing the risk of inconsistent approval decisions.

4.2 Relationship Between the ERD, Data Dictionary, and Assignment 1 Requirements

The ERD and data dictionary operationalise the functional requirements documented in Assignment 1. Requirement FR-01 (Digital Check-In/Check-Out) is supported by the WorkSchedule entity, which records the approved location for each day and enables the check-in process to validate whether the employee is at their assigned location. FR-03 (Notification Engine) is realised through the Notification entity, which stores every system-generated message triggered by request approvals and rejections. Non-functional requirement NFR-01 (99.5% uptime) influenced the data dictionary design by mandating that all key validation rules (such as the check that ReviewDate follows SubmissionDate) be enforced at the database level rather than solely in application logic, reducing the risk of data corruption during brief outages. The data dictionary provides the implementation-level detail — data types, key constraints, and validation rules — that transforms the conceptual entities from the ERD into a precise database schema specification.

4.3 Relationship Between This Assignment's Models and Assignment 2's Process Models

The models developed here integrate directly with the process models from Assignment 2. The HybridWorkRequest entity corresponds to data flows entering Process 5.0 (Manage Users & Config), where policy rules are maintained, and Process 4.0 (Send Notifications), which is triggered whenever a request reaches a final status. The WorkSchedule entity feeds into the Attendance Database (D1), which Process 1.0 (Manage Attendance) reads during check-in to validate the employee's expected work location. The Department entity aligns with the Policy & Config Database (D4), storing the 50% on-site minimum rule referenced by the context diagram's external entities. Together, these models form a consistent chain: the context diagram and DFDs established *what* the system does, the decision table specifies *how* the approval logic works, and the ERD with data dictionary defines *what data* the system needs to store to support those processes.

References

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